# SIGNAL & DATA ANALYSIS IN NEUROSCIENCE 2020

## NEURAL ENCODING

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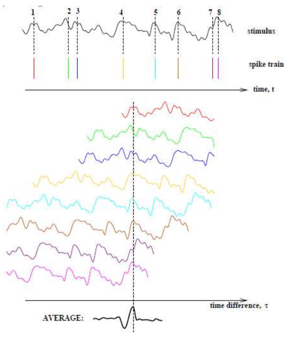
## Spike Triggered Average

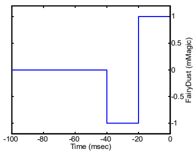
Provides information on average activity around (before and after) a spike

- Align sequence (e.g. LFP) around spike times (Can be the same sequence aligned at different times where spikes occurred)
- Average all instances of the activity around the spike (e.g. +/- 200ms)




## Example: (Exam 2006)

The fairy's 7th sense receptors respond to the appearance of fairy dust (measured in mMagic units) according to the following reverse correlation. Which of the following input A(t) will lead to maximal changes in the neuronal activity?




- Constant level of fairy dust (A=100).
- Wave of fairy dust, A(t)=100*sin(25*t*2π).
- Wave of fairy dust, A(t)=100*sin(50*t*2π).
- Wave of fairy dust, A(t)=100+50*sin(25*t*2π).
- Wave of fairy dust, A(t)=100+50*sin(50*t*2π).

## Linear Kernels (filters)

- Linear filter

$$r_{est} = r_0 + \int_0^\infty D(\tau)S(t-\tau)d\tau$$

*r0* – baseline firing rate
*D* - response kernel
*rest(t)* – firing rate function
*s(t)* – stimulus function

- The optimal kernel minimizes the err function defined below

$$Err = \int_0^T (r_{est}(t) - r(t))^2 dt$$

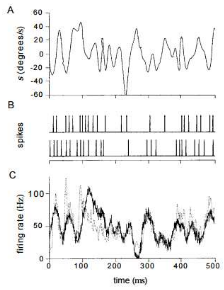
- The kernel is extracted from a stim with white noise distribution (The kernel is valid for every stimulus)

$$\boxed{D(\tau) = \frac{\langle r \rangle C(\tau)}{\sigma_s^2}}$$

$C(\tau)$ – STA $\langle r \rangle$ - average firing rate $\sigma_s^2$ - variance of stim

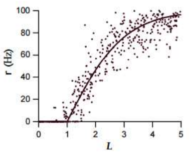
## Kernel estimation

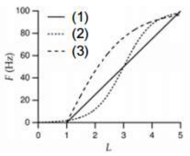
- Create white noise stimulus (no correlations).
- Calculate spike triggered average response to the stimulating white noise.
- Normalize the spike triggered average by the rate to get the optimal kernel.
- Estimate rate *rest(t)* assuming linear kernel




## Static non-Linearity

Neurons deviate from the linear relationship, mostly due to **saturation** and **threshold**.



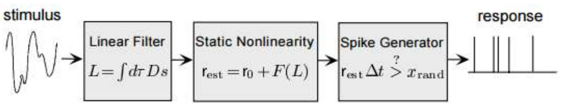

(1) $F(L) = G[L - L_0]_+$ -Rectifier

(2) $F(L) = \frac{r_{max}}{1 + \exp(g_1(L_{1/2} - L))}$ -Sigmoid

(3) $F(L) = r_{max}[\tanh(g_2(L - L_0))]_+$ -Hyperbolic

7

## Simulating spiking responses to stimuli




A model of the spike trains evoked by a stimulus can be constructed by using the firing rate estimate to drive a Poisson spike generator.